

Dynamic Programming

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Introduction

- ❑ Dynamic programming is a computational method that is used to align two protein or nucleic acid sequences.
- ❑ The method is very important for sequence analysis because it provides the very best or optimal alignment between sequences
- ❑ It provides optimal alignment by matching two sequences for all possible pairs of characters between the two sequences.
- ❑ This alignment will include matched and mismatched characters and gaps in the two sequences that are positioned so that the number of matches between identical or related characters is the maximum possible.
- ❑ Scoring Scheme: (e.g.,) Matches = +1, Mismatches = -1, Gap penalty = -2

A T G C C T

T G C C T

Matches = $1 \times (+1) = 1$

Mismatches = $4 \times (-1) = -4$

Gap penalty = $1 \times (-2) = -2$

Total score = $1 - 4 - 2 = -5$

A T G C C T

A T _ C C T

Matches = $x \times (+1) =$

Mismatches = $x \times (-1) =$

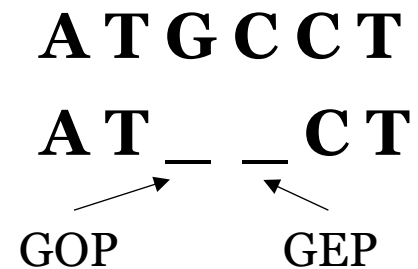
Gap penalty = $x \times (-2) =$

Total score = =

GOP & GEP

Gap open penalty: The penalty incurred when a new gap is introduced in the alignment

Gap extension penalty: The penalty applied for extending an existing gap (adding additional gaps).



AA sequences: Scoring matrices & gap penalties used to find optimal alignment

sequence 1	V	D	S	-	C	Y
sequence 2	V	E	S	L	C	Y
SCORE	4	2	4	-11	9	7
(26)						

Scoring Matrices & Gap Penalties

- ❑ Is the investigator interested in trying to find out whether two proteins have similar domains or structural features, whether they are in the same family with a related biological function, or whether they share a common ancestor relationship? The desired objective will influence the way the analysis is done.
- ❑ There are several decisions to be made along the way, including the type of program, whether to produce a global or local alignment, the type of scoring matrix, and the value of the gap penalties to be used.
- ❑ The Dayhoff PAM matrices, are based on an evolutionary model of protein change, whereas others, such as the BLOSUM matrices, are designed to identify members of the same family.
- ❑ Based on experience & what is suitable for alignment, the gap penalty and scoring matrices are decided.

Importance of Optimal Alignment

- ❑ Optimal alignments provide useful information to biologists concerning sequence relationships by giving the best possible information as to which characters in a sequence should be in the same column in an alignment, and which are insertions in one of the sequences (or deletions on the other).
- ❑ This information is important for making functional, structural, and evolutionary predictions on the basis of sequence alignments. Both global and local types of alignments may be made by simple changes in the basic dynamic programming algorithm.
- ❑ A global alignment program is based on the Needleman-Wunsch algorithm, and a local alignment program on the Smith-Waterman algorithm.
- ❑ Dynamic programming has also been used to perform multiple sequence alignment, but only for a **small number of sequences** because the complexity of the calculations increases substantially for more than two sequences.

Creation of alignment grid

- ❑ It is fundamentally similar to the dot matrix method in that it also creates a two dimensional alignment grid.
- ❑ However, it finds alignment in a more quantitative way by converting a dot matrix into a scoring matrix to account for matches and mismatches between sequences.
- ❑ By searching for the set of highest scores in this matrix, the best alignment can be accurately obtained.
- ❑ Dynamic programming works by first constructing a two-dimensional matrix whose axes are the two sequences to be compared.
- ❑ The residue matching is according to a particular scoring matrix.
- ❑ The scores are calculated one row at a time.
- ❑ This starts with the first row of one sequence, which is used to scan through the entire length of the other sequence, followed by scanning of the second row.
- ❑ The matching scores are calculated.

Creation of alignment grid

- ❑ The scanning of the second row takes into account the scores already obtained in the first round.
- ❑ The best score is put into the bottom right corner of an intermediate matrix.
- ❑ This process is iterated until values for all the cells are filled.
- ❑ Thus, the scores are accumulated along the diagonal going from the upper left corner to the lower right corner.
- ❑ Once the scores have been accumulated in matrix, the next step is to find the path that represents the optimal alignment.
- ❑ The best matching path is the one that has the maximum total score.